TEACHER GUIDE

AQA A-level Business (7138)

Operations management: scale



3.2 Unit: Managing people and operations
3.2.2 Operations management: scale

A revision lesson. Pupils run a car dealership in firms of four and revise operations management: scale on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same.

1 Unit cost per car

Round 1

A month of sales.

Unit cost: cost per car sold?

Fixed costs, a month	**£10,800**
Variable costs, a month	**£298,260**
Cars sold a month	**18 cars**

ANSWER

£17,170

(£10,800 + £298,260) ÷ 18 = £17,170

Why: Unit cost is all the costs shared between the cars.

2 Why unit cost rose

Round 1

After a takeover, your cost per car rose.

Which two diseconomies are most likely?

- **A** Poor communication across sites
- **B** Staff less motivated
- **C** Cheaper finance
- **D** Fixed costs spread wider
- **E** Bulk-buying discounts

ANSWER

✓ Poor communication across sites
✓ Staff less motivated

Why: Big firms can lose control and morale; the other three cut costs.

3 Rank unit cost fall

Round 2

Three plans and what they do to cost per car.

Order by % fall in cost per car, biggest first?

- **A** £2,000 down to £1,500
- **B** £8,000 down to £6,800
- **C** £5,000 down to £4,000

ANSWER

1. £2,000 down to £1,500
2. £5,000 down to £4,000
3. £8,000 down to £6,800

Why: Falls of 25%, 20%, 15%: the biggest fall in pounds is the smallest %.

Topic tasks

4 Unit cost, doubled

Round 2

Selling twice as many.

Cost per car if sales double?

Fixed costs, a month	**£16,000**
Variable cost per car	**£13,550**
Cars sold now	**16 cars**

ANSWER

£14,050

£16,000 ÷ 32 + £13,550 = £14,050

Why: The fixed costs are shared by 32 cars, not 16.

5 Stop growing?

Round 3

Each new branch cut your cost per car less, and the last one raised it.

What should you do next, and why?

A Stop growing for now

B Keep growing

ANSWER

✔ **Stop growing for now**

Why: Rising cost per car is a sign of diseconomies of scale.

6 Fixed cost spread %

Round 3

Spread the fixed costs.

Fall in cost per car, to the nearest %?

Fixed costs, a month	**£34,020**
Variable cost per car	**£270**
Cars sold now	**14 cars**
Cars sold after growth	**18 cars**

ANSWER

20%

£2,700 → £2,160; fall 20%

Why: The fixed costs are shared by more cars after growth.

Final questions

Each pupil answers 6 final questions on operations management: scale. They come from these 8 questions.

1. A group buys the next town's family dealership, keeping its staff and site. Which type of growth is this, and its main risk?
   ✔ **External, and the risk is merging two ways of working**

2. A dealer group's total costs are £2,400,000 for 600 cars sold. After growth they are £3,500,000 for 1,000 cars. By how much does unit cost fall?
   ✔ **£500**

3. A group of ten dealerships orders tyres together and gets a lower price per tyre. Which economy of scale is this?
   ✔ **Purchasing economies**

4. A large dealer group borrows at a lower interest rate than a single-site dealer can. Which economy of scale is this?
   ✔ **Financial economies**

5. After rapid growth, decisions take weeks and branch staff feel ignored by head office. What is this an example of?
   ✔ **Diseconomies of scale from communication and coordination**

6. A dealer opens a second showroom with its own profits and its own staff. Which type of growth is this?
   ✔ **Organic (internal) growth**

7. Two rival dealer groups in the same region combine into one company. What type of growth is this?
   ✔ **Horizontal integration**

8. A dealer group buys the company that supplies its parts. What is the main gain from this backward vertical integration?
   ✔ **More control over supply and its cost**